



Chiara De Luca

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SUMMARY

As a dedicated neuroinformatics researcher, I am deeply passionate about unraveling the complexities of the brain. My work centers on developing bio-inspired spiking neural networks and implementing them on neuromorphic chips, aiming to create energy-efficient, intelligent machines that emulate the brain's learning and adaptive capabilities. With a robust interdisciplinary background in physics and neuroscience, I bring a unique perspective to advancing the field of artificial intelligence, striving to make impactful contributions that bridge the gap between biological systems and technology.

EMPLOYMENT

Jun 2024 - ongoing	Postdoctoral researcher Digital Society Initiative, University of Zurich PI of the founded project “Neufarm” (grant no.G-95017-01-12) ZNZ junior group leader
Jun 2023 - Jun 2024	Postdoctoral researcher Institute of Neuroinformatics, University of Zurich and ETH Zurich Supervisors: Prof. Matthew Cook; Prof Giacomo Indiveri

EDUCATION

Nov 2019 - May 2023	PhD (Behavioural Neuroscience) at La Sapienza University of Rome <i>cum honour</i> Thesis title: “ <i>Computational approaches to neuronal dynamics. Analysis and model of neuronal activity through spiking simulations</i> ” Supervisor: Prof. Stefano Ferraina, Prof. Pier Stanislao Paolucci
Nov 2017 - Sep 2019	MSc in Physics at La Sapienza University of Rome 110/110 <i>cum laude</i> Thesis title: “ <i>Learning and sleep in a thalamo-cortical multi-area model</i> ” Supervisor: Prof. Pier Stanislao Paolucci
Nov 2014 - Jul 2017	BSc in Physics at La Sapienza University of Rome 110/110 Dissertation: “ <i>Monte Carlo Methods applied to the Ising model</i> ” Supervisor: Enzo Marinari, Full Professor

MENTORING

PhD Supervision

- PhD Lead Supervisor, June 2024 - ongoing, Candidate: Mirco Tincani
- PhD co-supervisor, Institute of Neuroinformatics, University of Zurich, ongoing.
Candidate: Leonarto Martinelli
- PhD co-supervisor, Institute of Neuroinformatics, University of Zurich, ongoing.
Candidate: Tianshu Shen
- PhD co-supervisor, Institute of Neuroinformatics, University of Zurich, ongoing.
Candidate: Patrick Bösch
- PhD co-supervisor, Institute of Neuroinformatics, University of Zurich,
and Italian Institute of Technology (IIT), Genova, Italy, ongoing.
Candidate: Irene Ambrosini
- PhD co-supervisor, Italian Institute of Technology (IIT), Genova, Italy, ongoing.
Candidate: Bernard Maacaron

Master Thesis

- “Mixed-signal implementation of feedback-control optimizer for single-layer Spiking Neural Networks”,
Universität Stuttgart, September 2025
Candidate: Jonathan Haag
- “Modeling Proprioceptive Representation of Hand Kinematics on DYNAP-SE2”, ETH Zurich, August
2025
Candidate: Benjamin Horner
- “Neuromorphic twin of somatosensory cortex: an on-chip network with bio-plausible adaptive re-
sponses”, ETH Zurich, November 2024
Candidate: Tianshu Shen
- “A Hardware-Aware Spiking Autoencoder for Spatio-Temporal Signal Processing”, ETH Zurich, Octo-
ber 2024
Candidate: Patrick Bösch
- “Spike-timing dependent bio-inspired algorithms on analogue neuromorphic architectures”, Sapienza
University of Rome, September 2023
Candidate: Luigi Rosati
- “Thalamo-cortical model of multi-layer spiking neural network capable of learning and sleep”, Sapienza
University of Rome, January 2021 Candidate: Leonardo Tonielli
- “Learning of handwritten digit classes, their position and velocity in a thalamo-cortical multi-scale
spiking network”, Sapienza University of Rome, July 2020
Candidate: Davide Cipollini

Bachelor Thesis

- ‘On-Chip Spiking Neural Network for PPG-Based Blood Pressure Estimation’, ETH Zurich, September
2026
Candidate: Mazlum Oyman
- “Time-Invariant Shape Recognition Using Tactile Sensors on Neuromorphic Hardware”, , ETH Zurich,
September 2025
Candidate: Yannis Hernandez
- “A neuromorphic continuous soil monitoring system for precision irrigation”, ETH Zurich, January
2025
Candidate: Khaled Kerouch

Semester Projects

- “Hybrid R-AdEx Neuron Model”, ETH Zurich, June 2026
Candidate: Ludovica Gandolfi
- “Nebula Defender - A Cochlear-Inspired Spiking Neural Network for Real-Time SSVEP Steering and Blink-to-Shoot Control”, ETH Zurich, June 2026
Candidate: Edoardo Ghio
- “Neuromorphic Heart Rate Estimation from Wrist-Worn Accelerometer During Sleep”, ETH Zurich, May 2026
Candidate: Marvin Jarju
- “Toward Multi-Input ICMS: Scaling a Neuromorphic Model of Somatosensory Cortex”, ETH Zurich, May 2026
Candidate: Mar Tineo Rubio
- “Training Neuromorphic Devices using Novel Spiking Feedback Algorithm”, ETH Zurich, April 2025
Candidate: Christian Metzner

TEACHING

- Neuromorphic Intelligence Spring semester 2026
- DSI Minor module Digital Transformation - a Scientific Overview Fall semester 2024
- Mentor in the AMGHEN Program June-August 2025-2026

TALKS AND AWARDS

Invited talks

- ECCV NeVi Workshop, Malmo, 9th September 2026
- ICDL NCDL Workshop, Prague, 16th September 2025
- World Life Science Conference (WLSC24), China October 2024
- NetZero Conference, United Kindom, September 2024
- CNR Retreat IN, Italy, September 2022

Awards

- Best poster awards. Nature Forum: The Future of Sensing Technologies Apr 2026
- Best poster award. HBP Summit Mar 2021
- Best poster and presentation awards. HBP-Student Conference Dec 2021
- Best poster and presentation awards. HBP-Student Conference Jan 2020

Grants

- DSI Bridge Postdoc Call (G-95017-01-12) Jan 2024
The DSI bridge postdoc positions address researchers at the outset of their academic career, carrying out interdisciplinary and inter-institutional research on topics closely linked to digitalization and its impact on society.
- COST Action Proposal (OC-2025-1-28705) July 2025
”Pan-European network on neuromorphic computing - from neuromorphic devices to brain-inspired neuromorphic computing

Organized events

- CapoCaccia Neuromorphic Workshop, organizing committee 2025 - ongoing
- Bio-inspired DL Workshop, Germany June 2026
- IEEE Conference on AgriFood Electronic (CAFE),
technical program committee September 2026
- International Conference on Neuromorphic Systems (ICONS),
organizing committee August 2024-2025
- BASSES Workshop, Rome. Italy. June 2022

PUBLISHED ARTICLES

- Ambrosini, Irene et al. (2026). “Training slow silicon neurons to control extremely fast robots with spiking reinforcement learning”. In: *2026 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, pp. 489–493.
- D’Angelo, Giulia et al. (2026). “A benchmarking framework for embodied neuromorphic agents”. In: *Nature Machine Intelligence* 8.3, pp. 300–312.
- Ferrari, Federica et al. (2026). “Neuromorphic Spiking Ring Attractor for Proprioceptive Joint-State Estimation”. In: *2026 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, pp. 4162–4166.
- Haag, Jonathan et al. (2026). “Mixed-signal implementation of feedback-control optimizer for single-layer Spiking Neural Networks”. In: *2026 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, pp. 3146–3150.
- Martinelli, Leonardo et al. (2026). “Hardware-Aware Attractor Dynamics in Neuromorphic Systems: Compensating for Device Mismatch through Network Design”. In: *2026 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, pp. 3151–3155.
- Costa, Filippo and Chiara De Luca (2025). “Continuous signal sparse encoding using analog neuromorphic variability”. In: *Neuromorphic Computing and Engineering*.
- De Luca, Chiara and Elisa Donati (2025). “Queen Detection in Beehives via Environmental Sensor Fusion for Low-Power Edge Computing”. In: *2025 IEEE Conference on AgriFood Electronics (CAFE)*. IEEE, pp. 1–5.
- De Luca, Chiara, Mirco Tincani, et al. (2025). “A neuromorphic multi-scale approach for real-time heart rate and state detection”. In: *npj Unconventional Computing* 2.1, p. 6.
- Saponati, Matteo et al. (2025). “A feedback control optimizer for online and hardware-aware training of Spiking Neural Networks”. In: *2025 Neuro Inspired Computational Elements (NICE)*. IEEE, pp. 1–10.
- Tincani, Mirco et al. (2025). “A neuromorphic continuous soil monitoring system for precision irrigation”. In: *2025 IEEE Conference on AgriFood Electronics (CAFE)*. IEEE, pp. 1–5.
- Carpegna, Alessio et al. (2024). “Neuromorphic Heart Rate Monitors: Neural State Machines for Monotonic Change Detection”. In: *2024 IEEE Biomedical Circuits and Systems Conference (BioCAS)*. IEEE, pp. 1–5.
- Gomez, Maria Engracia et al. (2024). “RED-SEA Project: Towards a new-generation European interconnect”. In: *Microprocessors and Microsystems* 110, p. 105102.

- Gutzen, Robin et al. (2024). “A modular and adaptable analysis pipeline to compare slow cerebral rhythms across heterogeneous datasets”. In: *Cell Reports Methods* 4.1.
- De Luca, Chiara, Cristiano Capone, et al. (2023). “Simulations approaching data: cortical slow waves in inferred models of the whole hemisphere of mouse”. In: *Communications biology* 6.1, p. 266.
- De Luca, Chiara, Bruno Golosio, et al. (2021). “Thalamo-cortical spiking model of incremental learning combining perception, context and NREM-sleep”. In: *PLoS Computational Biology* 17.6, e1009045.
- Golosio, Bruno et al. (2021). “Fast simulations of highly-connected spiking cortical models using GPUs”. In: *Frontiers in Computational Neuroscience* 15, p. 627620.
- De Luca, Chiara, Marco Celotto, et al. (2020). “Analysis and model of cortical slow waves acquired with optical techniques”. In: *Methods and protocols* 3.1, p. 14.